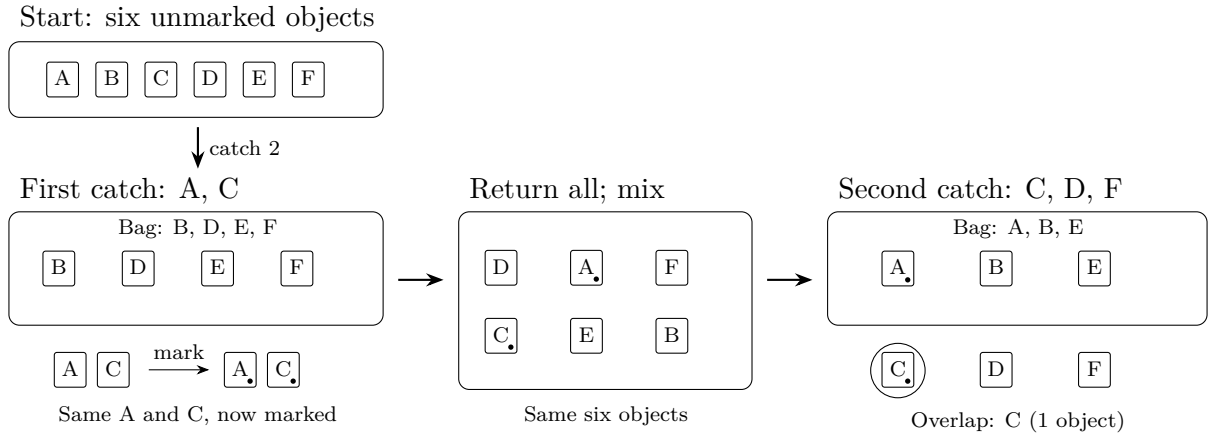
Use identical-size cards or counters with visible marker dots that cannot be felt. Keep the same objects throughout a series; add, remove or exchange none except for temporary catches. For sampling trials, draw without looking; every group of the chosen size should have the same chance, whether marked or unmarked.

Keep each catch out until it is counted. Mark the first catch, return *all* of it, and mix before the second catch. Marks stay on through both catches. An overlap is a marked object in the second catch. Before a new round, return everything and clear the marks. To show a record is possible, you may arrange catches: use distinct objects in each catch, from the same bag.



Problem 1: The keeper hides 12 to 24 objects. Catch 6, mark and return them, then catch 6 again. Play three rounds with the same objects. Choose a guess before the keeper reveals the total.

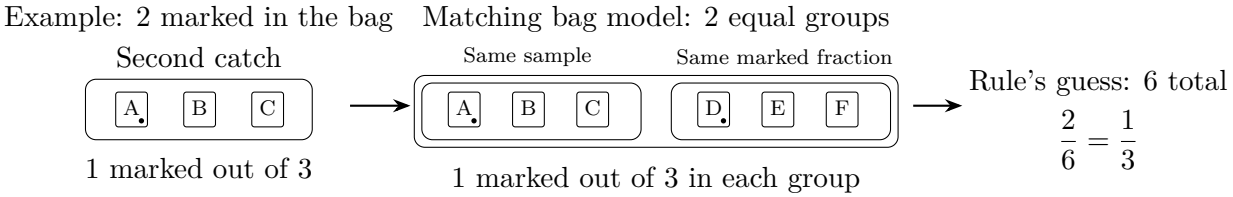
Round	Overlap	My guess
1		
2		
3		

Keeper's total: _____

Problem 2: Both records have first catch 6 and second catch 6. For each record, build the smallest possible bag. Is there a largest possible bag if you have as many cards as you want?

Record	Overlap	Smallest total
A	2	
B	0	

Problem 3: A proposed guessing rule chooses a total that makes the marked fraction in the bag match the marked fraction in the second catch.



Find the rule's guess for record A. Does the catch record force that total? Make a different possible bag to test your answer.

